



Be lazy, shift right

Arguments to reduce the cognitive load
in agent development





Embedding AI

Staying cool behind the bar of the AI
revolution



Agenda

- Introduction to challenges with AI
- Discuss # AI architectures
- Discuss challenges to bring AI to production
- Discuss AI dev workspace
- Demo

While reflecting on cog. load and shift right.



Disclaimer

- Everything is my own opinion; feel free to disagree.
- A lot of content: They are touchpoints to show you what you have to think about.
⇒ Sometimes dense slides, but then you can use them as a guide afterward.
- You are always free to reach out.



**“Keep your
options open!”**



Maarten Vandepierre

maarten-vandepierre.be





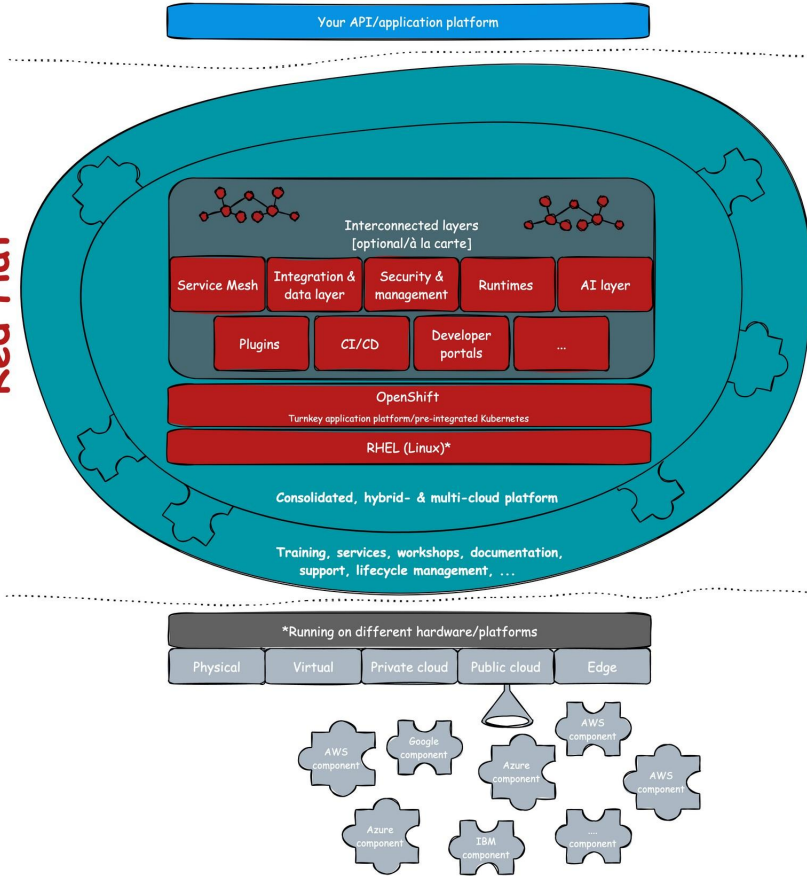
*What is
cognitive load?*

Be lazy, shift right

Arguments to reduce the cognitive load
in agent development



Red Hat



Applications

App & AI platform
⇒ software factory

Hybrid infrastructure

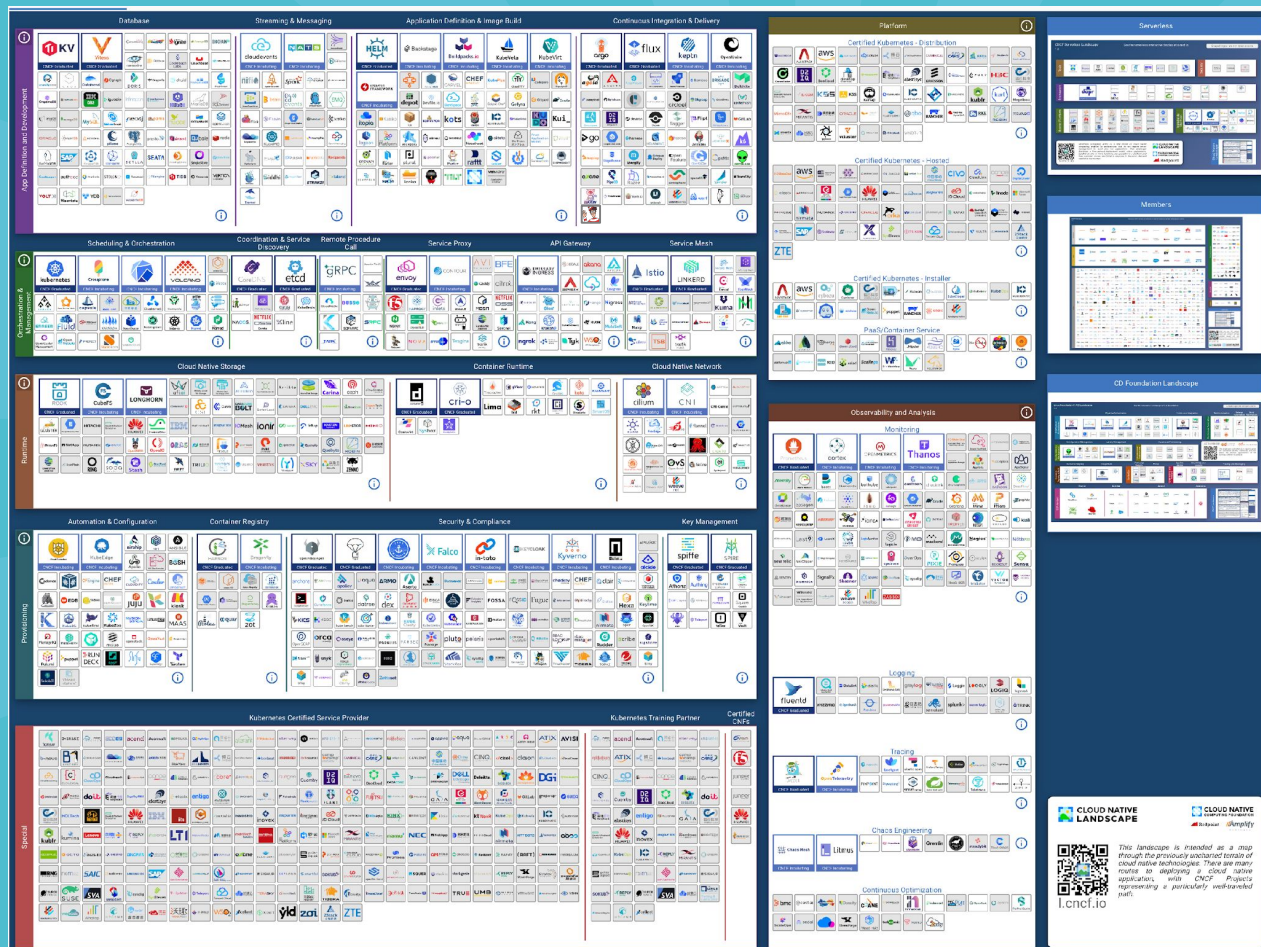
Cognitive load

The Cloud-Native Grocery Store

Where is

- Wally?
- Waldo?
- Charly?

Source: <https://landscape.cncf.io/>



Cognitive load

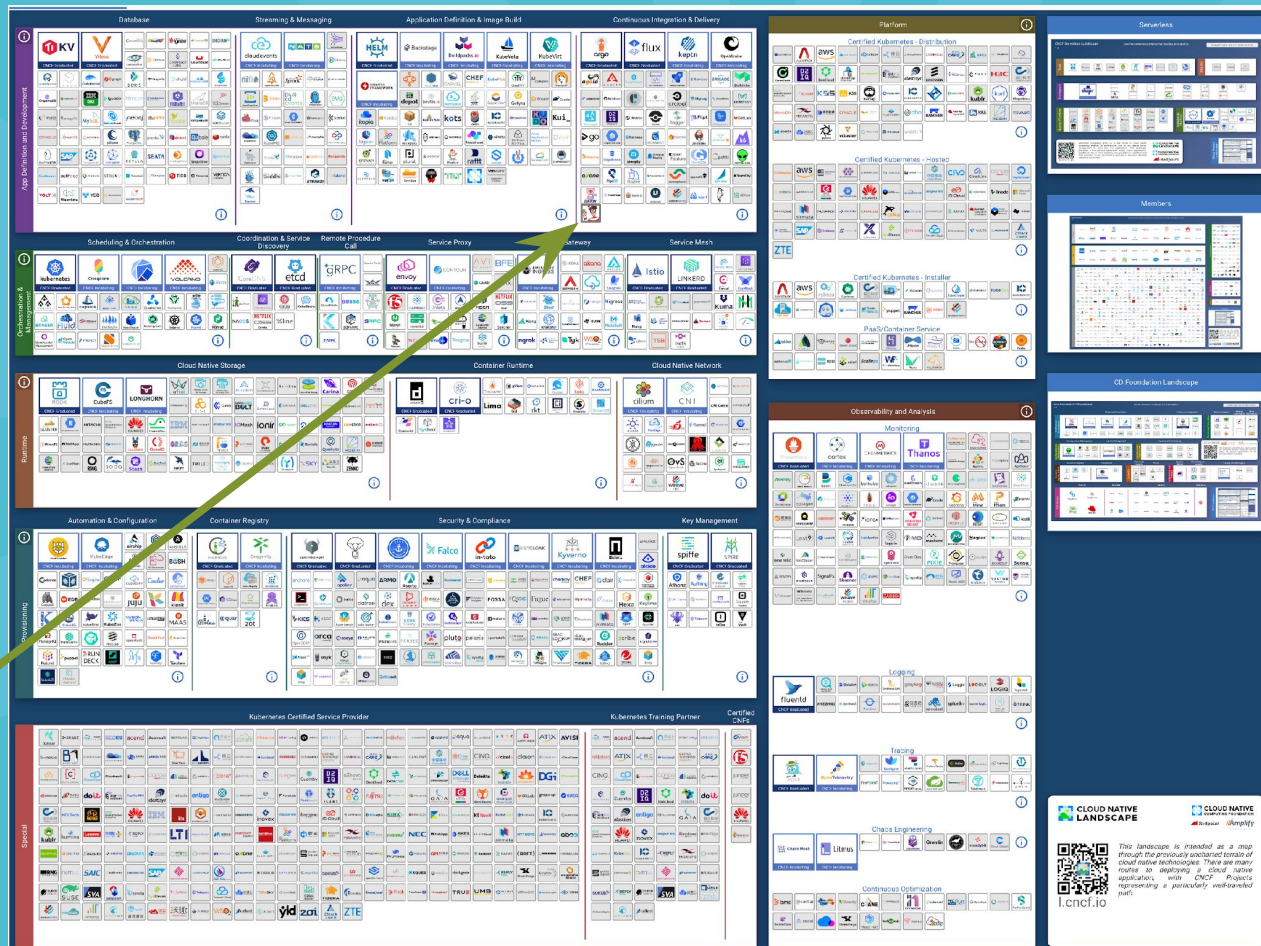
The Cloud-Native Grocery Store

Where is

- Wally?
- Waldo?
- Charly?



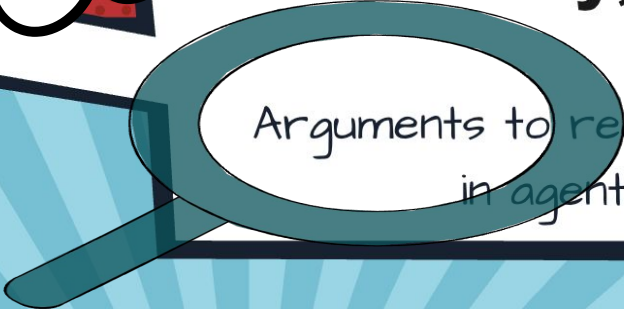
Source: <https://landscape.cncf.io/>





What are we going to solve?

Be lazy, shift right



Arguments to reduce the cognitive load
in agent development

A man with curly brown hair and glasses, wearing a dark jacket over a light-colored shirt, is looking at a smartphone. He has a brown backpack strap over his shoulder. In front of him is a stack of papers and a dark mug. A large, white, cloud-like thought bubble with a thick black outline is positioned above him, containing the text. The background is a vibrant red with numerous yellow and orange diagonal streaks radiating from the center, creating a sense of energy and movement.

Behind the bar of AI

What's really being served?

Real world news

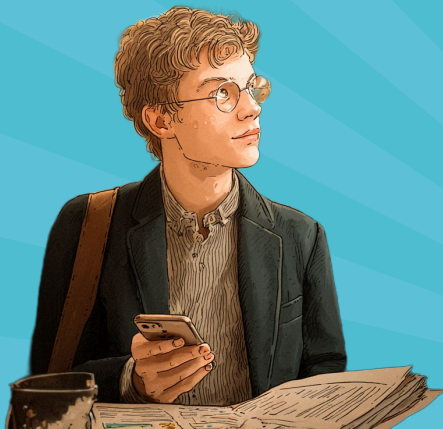
Cyber Security

What the CLOUD Act Really Means for EU Data Sovereignty

The CLOUD Act allows U.S. authorities to access data stored in the EU, putting it in direct conflict with GDPR. Learn how this impacts data sovereignty and what EU businesses can do to stay compliant

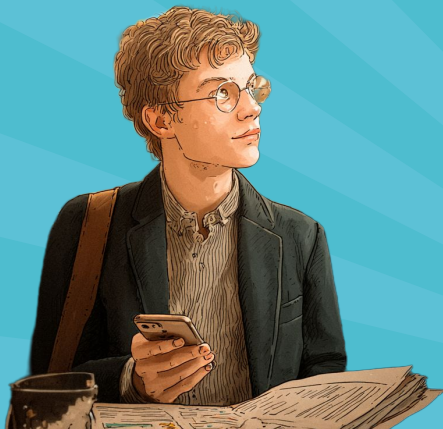


Wire
Jul 15, 2025



Source: <https://wire.com/en/blog/cloud-act-eu-data-sovereignty>

Real world news



EU data not protected from US access

21 July, 2025 11:35

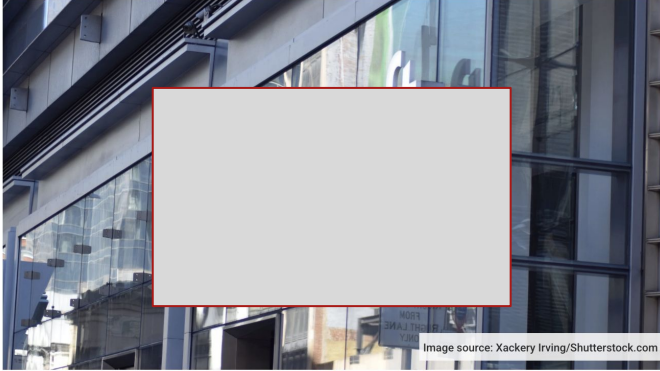


Image source: Xackery Irving/Shutterstock.com

Facebook

X

LinkedIn

Reddit

WhatsApp

Pocket

 This article is also available in German.

Explosive testimony before the French Senate: cannot guarantee that European user data is safe from US authorities despite EU hosting.

cannot protect European user data from access by US authorities – even if it is stored in EU data centers. This was admitted under oath by the legal director of France, , at a [hearing before the French Senate](#) on June 10, 2025.

Source: [https://www.it-daily.net/en/shortnews-en-eu-data-not-protected-from-us-access](https://www.it-daily.net/en/shortnews-en-<div></div>eu-data-not-protected-from-us-access)

Real world news

Daryna Antoniuk

June 13th, 2025

Technology

Government

News

Danish government agency to ditch Microsoft software in push for digital independence

Denmark's tech modernization agency plans to replace Microsoft products with open-source software to reduce dependence on U.S. tech firms.



In an interview with the local newspaper *Politiken*, Danish Minister for Digitalisation Caroline Stage Olsen confirmed that over half of the ministry's staff will switch from Microsoft Office to LibreOffice next month, with a full transition to open-source software by the end of the year.



Real world news

NEWSLETTERS · CFO DAILY

MIT report: 95% of generative AI pilots at companies are failing



BY SHERYL ESTRADA

SENIOR WRITER AND AUTHOR OF CFO DAILY

August 18, 2025 at 6:54 AM EDT



GETTY IMAGES

Forbes

EDITORS' PICK | BUSINESS > MANUFACTURING

Why 95% Of AI Pilots Fail, And What Business Leaders Should Do Instead

By [Andrea Hill](#), Contributor. © Andrea Hill is a multi-industry CEO covering bu...

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Published Aug 21, 2025, 11:54am EDT, Updated Aug 28, 2025, 07:54am EDT

Source: <https://www.forbes.com/sites/andreahill/2025/08/21/why-95-of-ai-pilots-fail-and-what-business-leaders-should-do-instead/>
<https://fortune.com/2025/08/18/mit-report-95-percent-generative-ai-pilots-at-companies-failing-cfo/>



Recap

- Challenges:
Cognitive load, what about the data, SDLC/WDLCs





*What is agentic,
RAG, MCP, ... ?*

Be lazy, shift

Arguments to reduce the cognitive load
in agent development



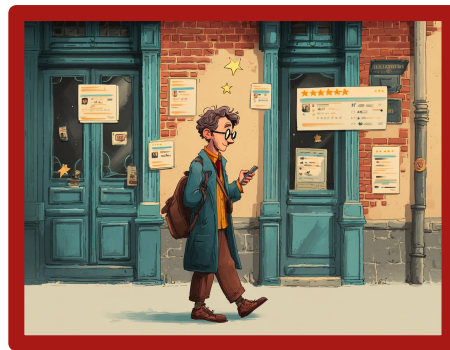
Three Friends going to a bar..



Fine-Tuna



Agent Gaga

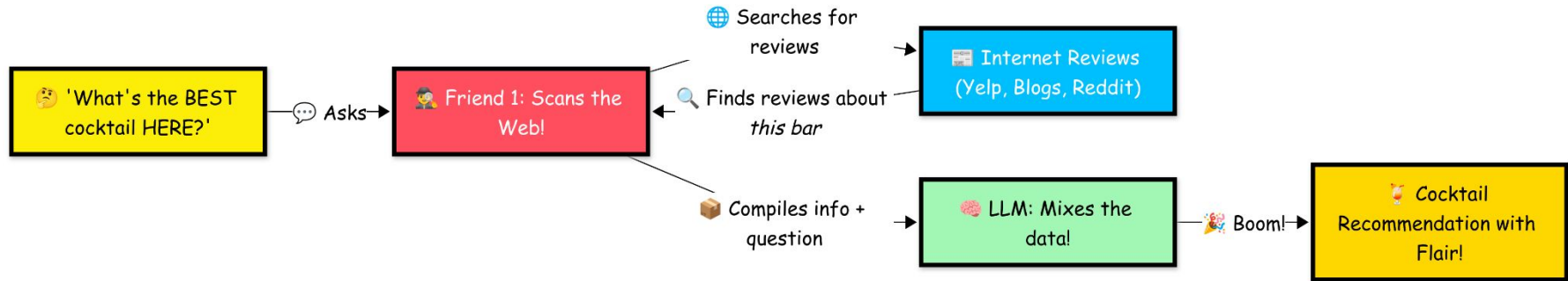


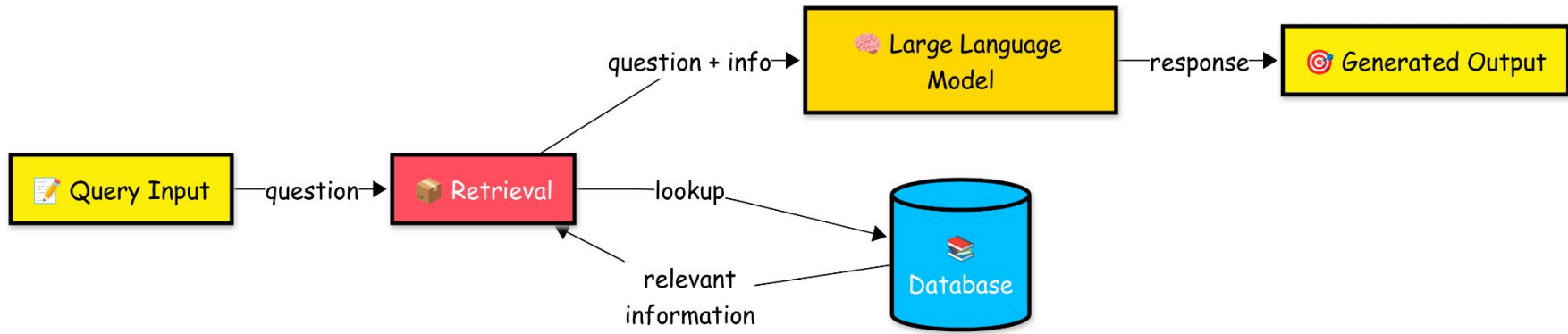
Retrievee

Meet Retrievee

- Immediately pulls out phone.
 - Orders based on a 4.3-star review from 2017.
- Knows nothing, but finds everything.







Meet Fine-Tuna

- Always orders the same thing.
 - Panics if the menu changed.
- Memory-based. No improvising.



RAG vs Fine-Tuning



	RAG	Fine-tuning
External Access	Yes	No
Response Time	Slow	Fast
Knowledge Update	Easy	Difficult
Adaptability	High	Lower
Inference Cost	Higher	Lower



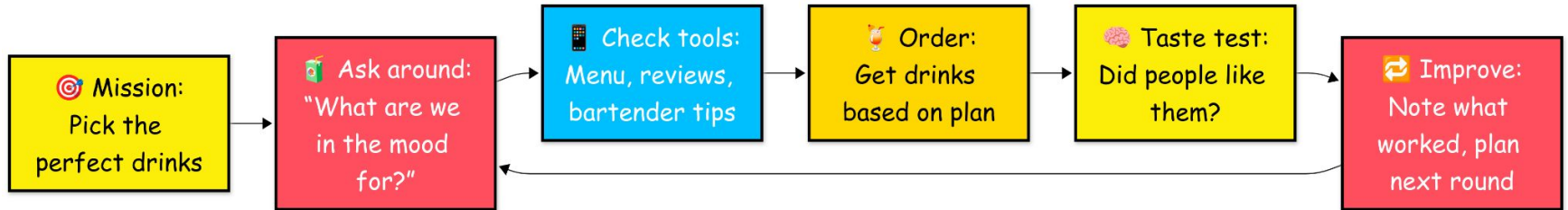
InstructLab

Meet Agent Gaga

- Already talked to the bartender.
- Pre-ordered your drinks.
- Tools: one language $\Rightarrow$ MCP

→ Combines memory, research, and vibes.





The risks

Lack of transparency
→ No clear accountability



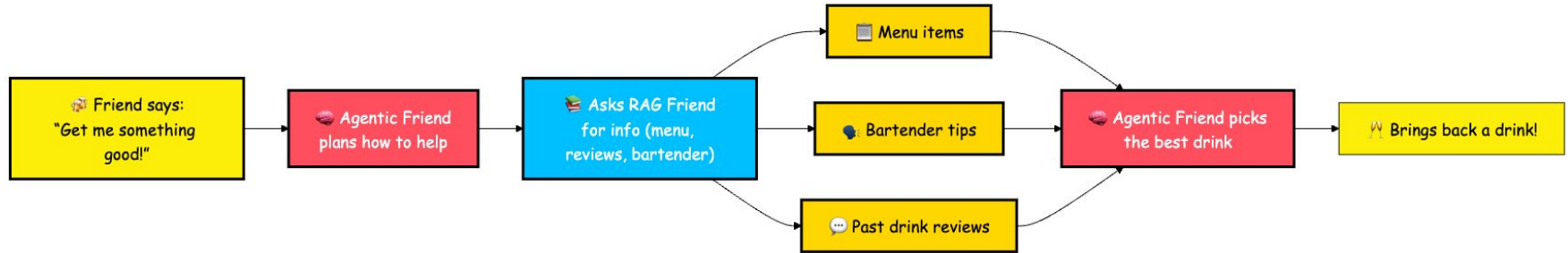
Agentic + RAG



+



AGENTIC RAG





Be lazy, shift right

*So, we want/need
"to do AI"*

to reduce the cognitive load
agent development



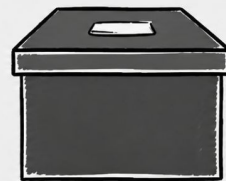


1: Query



Main application

2: Interact with AI



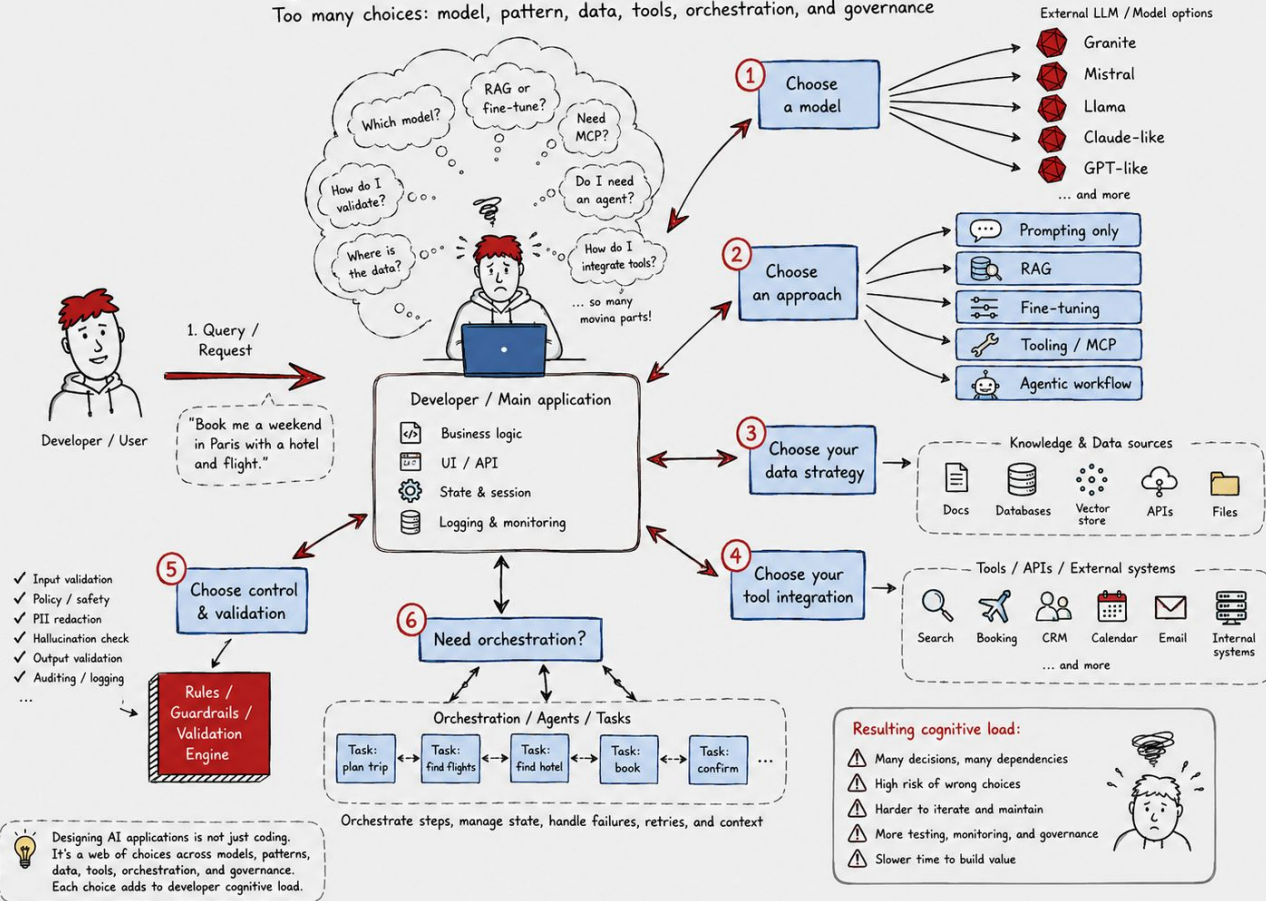
AI

*So... I'm doing AI,
right?*



Cognitive load for developers in AI application design

Too many choices: model, pattern, data, tools, orchestration, and governance

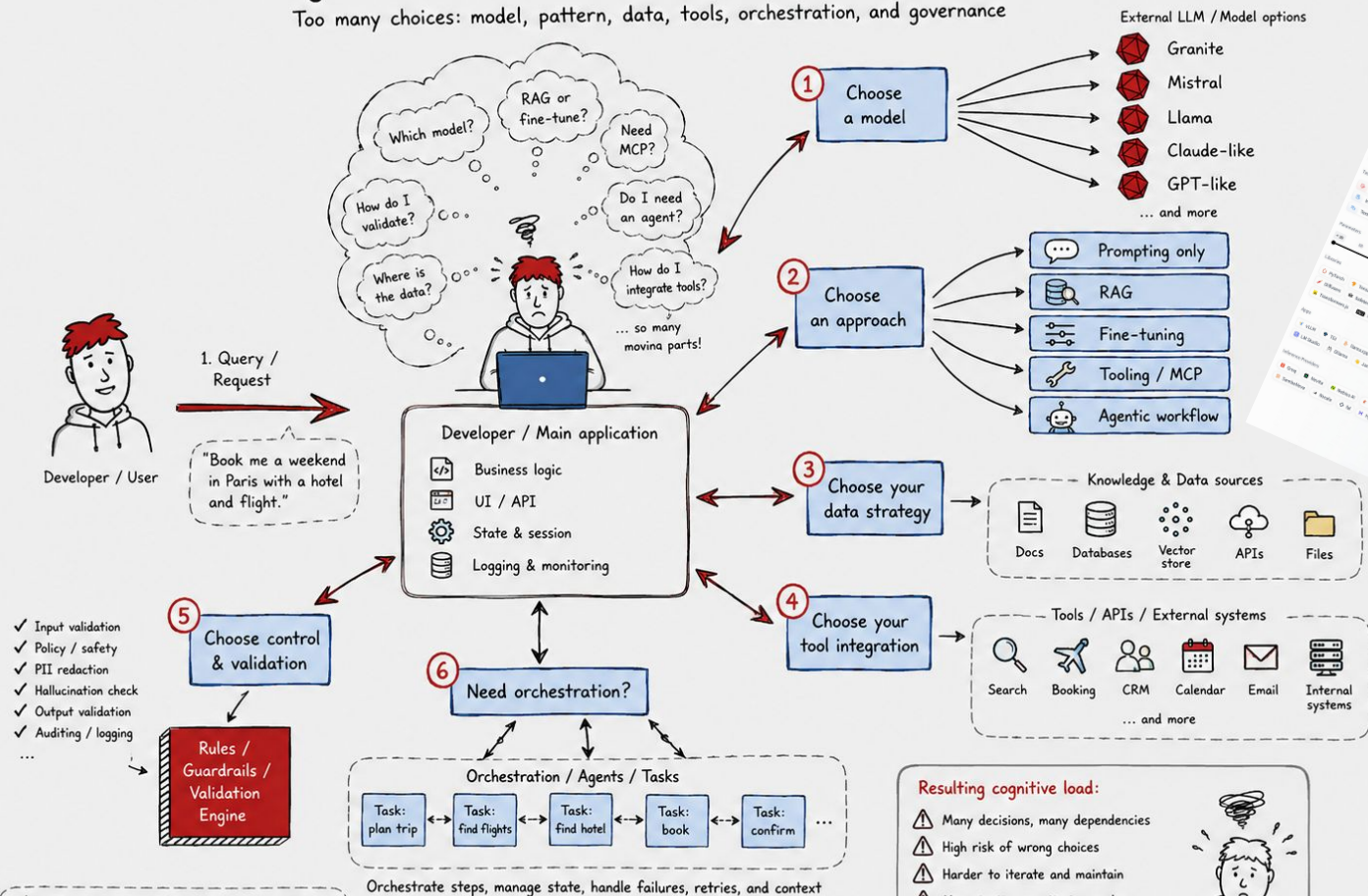


Owww.... f*ck...
is OpenAI, Anthropic,
model-as-a-service,...
fixing this?



Cognitive load for developers in AI application design

Too many choices: model, pattern, data, tools, orchestration, and governance



Designing AI applications is not just coding. It's a web of choices across models, patterns, data, tools, orchestration, and governance. Each choice adds to developer cognitive load.



Recap

- Challenges:
Cognitive load, what about the data, SDLC/WDLCs
- # AI architectures/implementations
- "Just Calling AI"

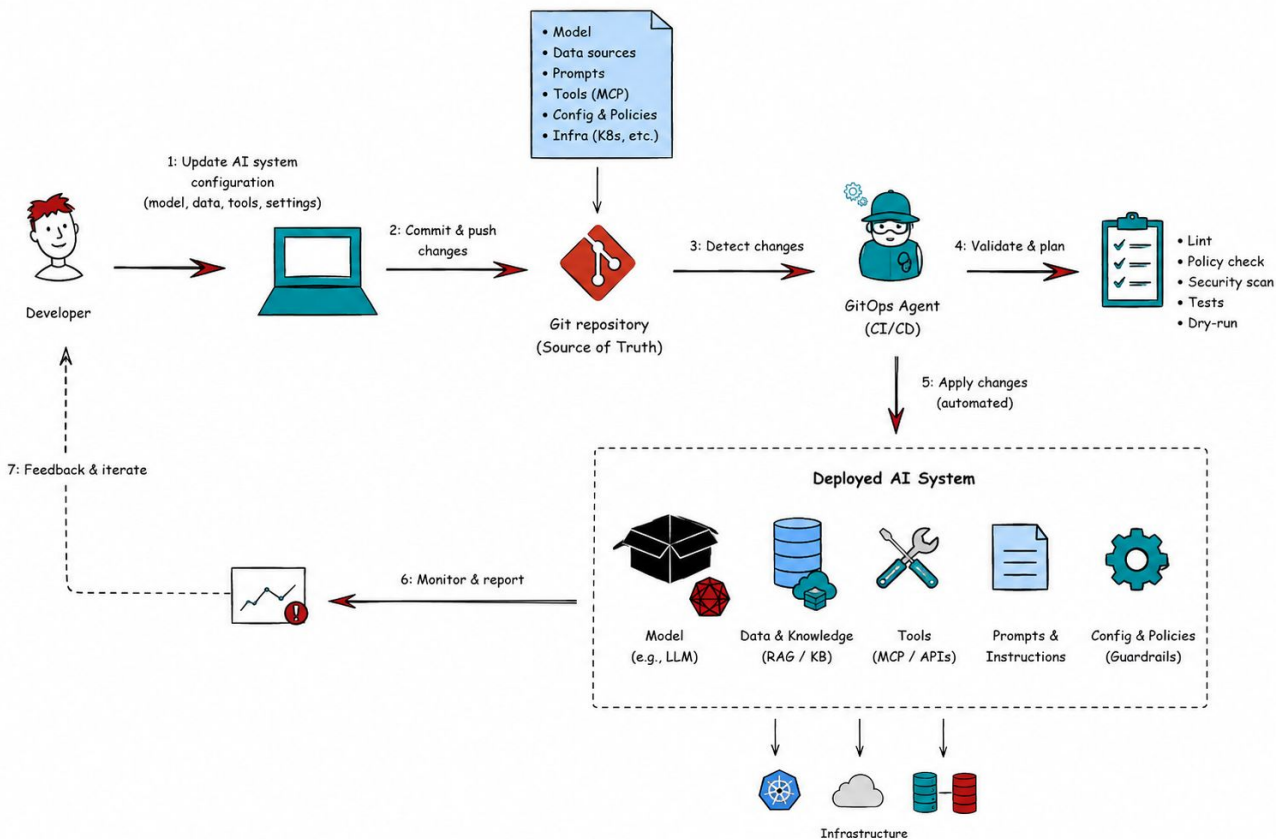




Be lazy, shift right

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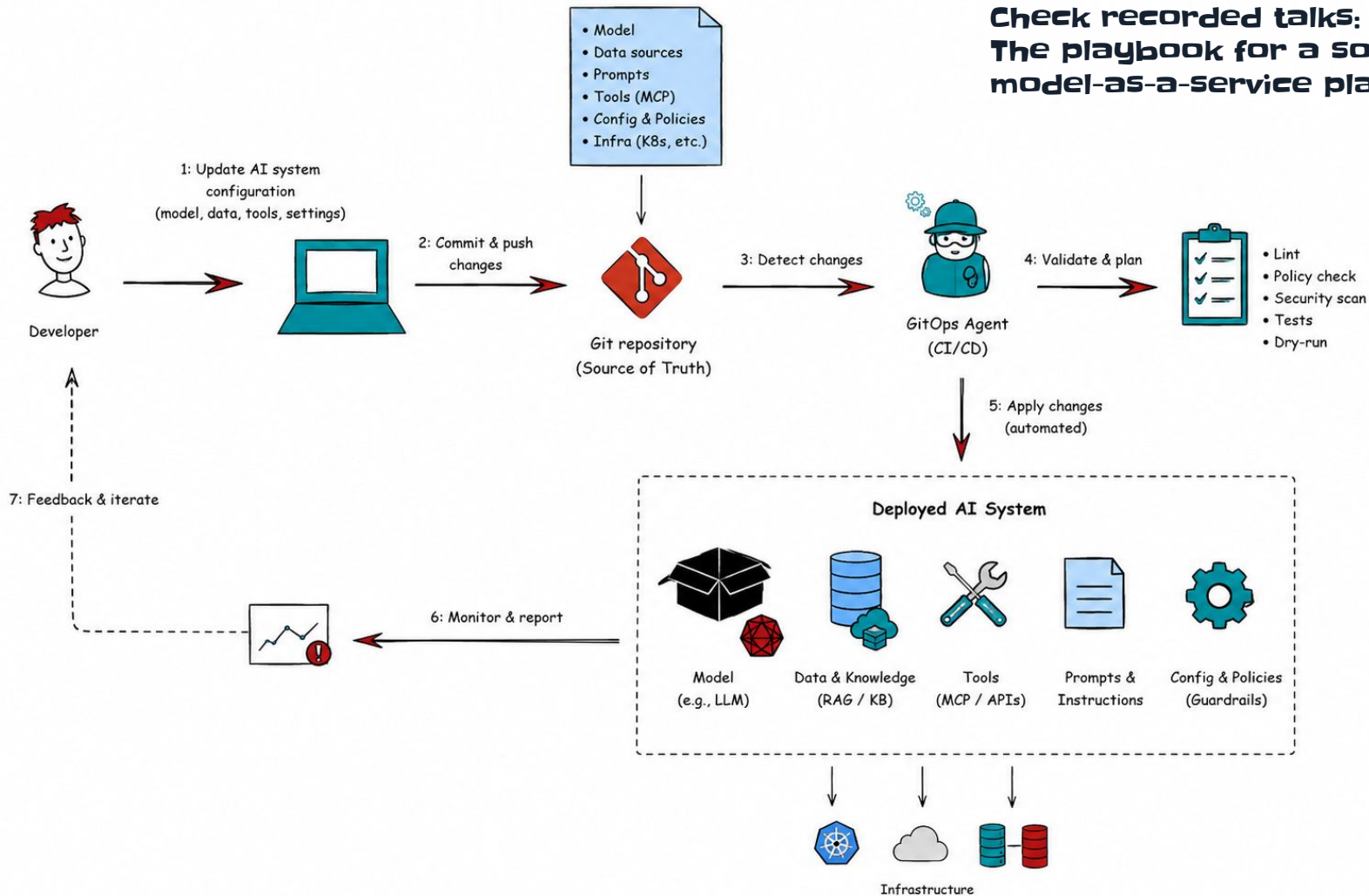
*F*ck, we need
to deploy this*



So.... How to
deploy all of
this? Team or
individual?



Check recorded talks: The playbook for a sovereign model-as-a-service platform.



Check blogs:
The subject matter expert
advantage in the AI era



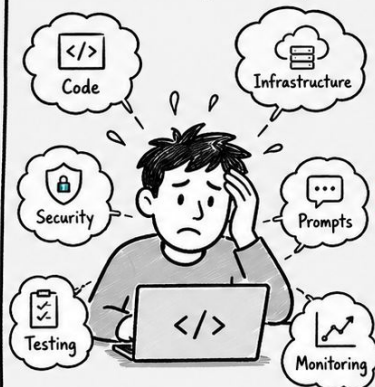
AI delivery is a team initiative



Responsibility shifts from one individual to a human-and-agent team.

OLD VIEW

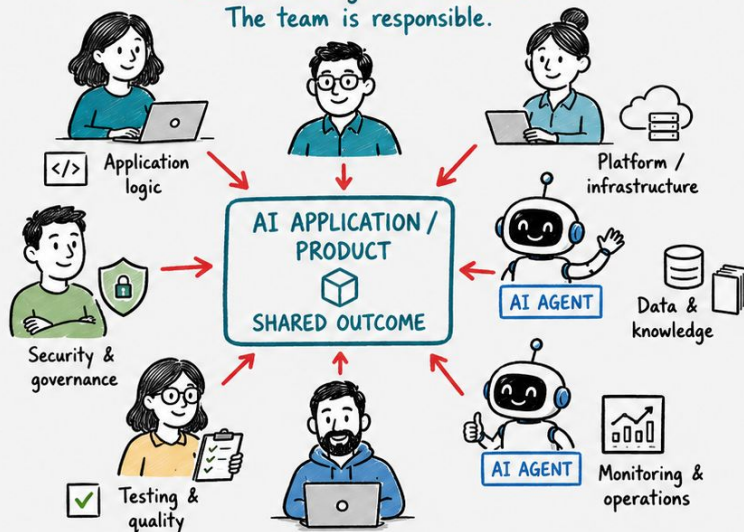
Individual responsibility



Too much responsibility
on one person

TEAM RESPONSIBILITY

Humans + AI agents collaborate.
The team is responsible.



THE RESULT:



Shared
ownership



Reduced
individual
pressure



Better
collaboration



More
resilient
delivery

So... Who is responsible?
Moving from
individual ownership to
team ownership

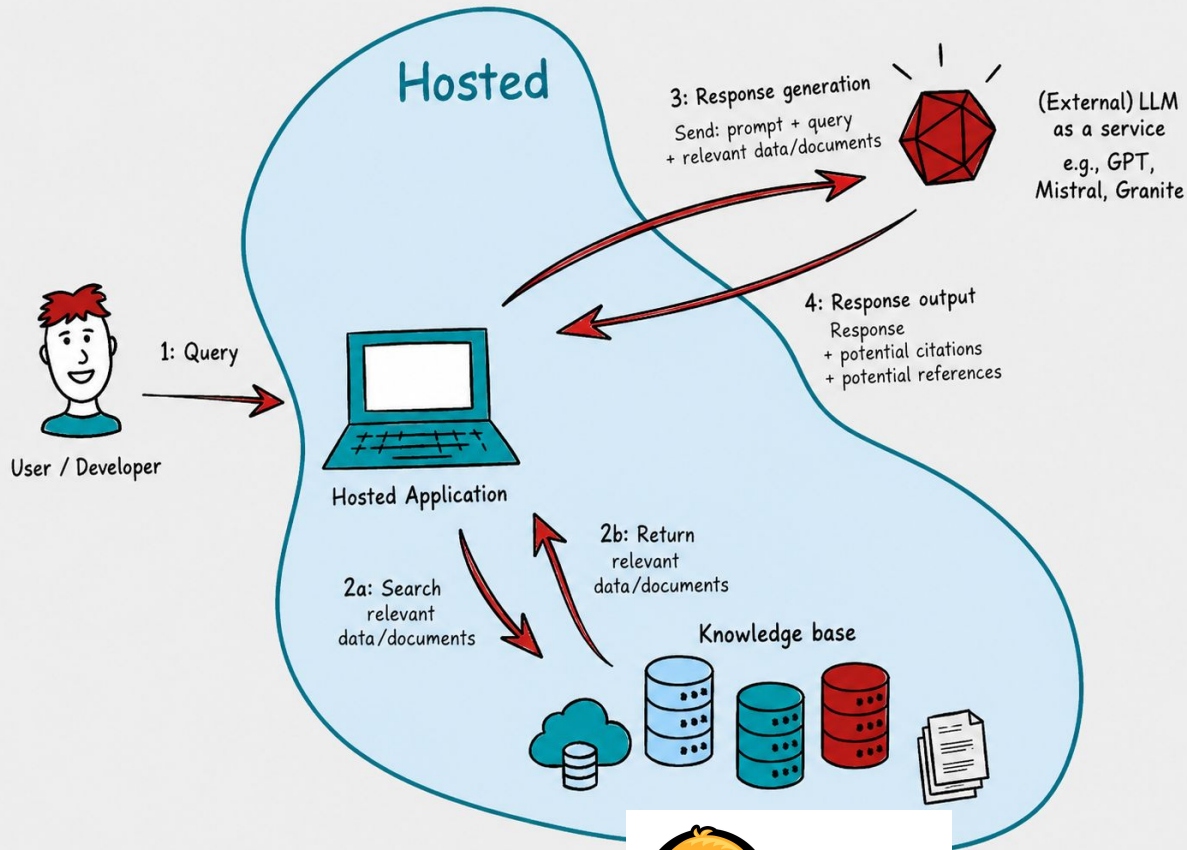




Be lazy, shift right

Arguments to reduce the cognitive load
in agent development

*They need data.
How do we pass it?*

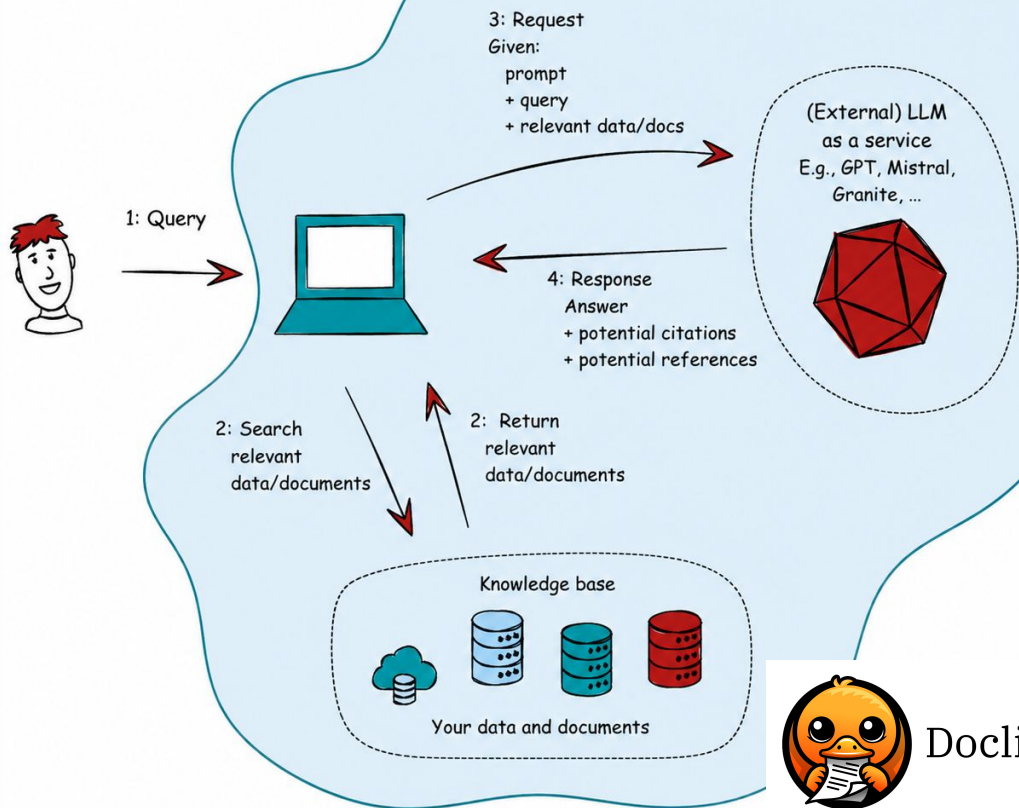


Docling

So... database is
on-prem... We're
good, right?
⇒ Speaking in the pub



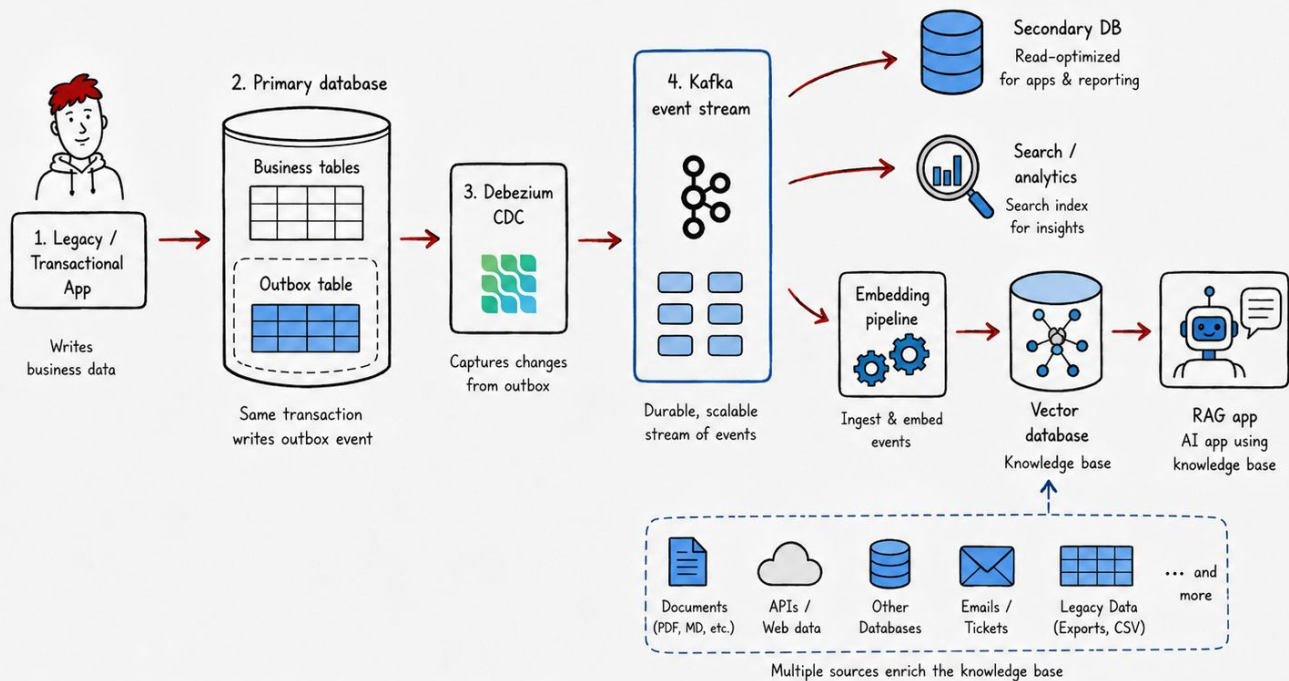
Hosted



Docling

*So... some discussions
and/or secrets should
not be heard by the
barkeeper, other
guests, ...*





Why this helps modernization

- Decouple legacy writes from new reads
- Build new services safely
- Reuse events for multiple consumers

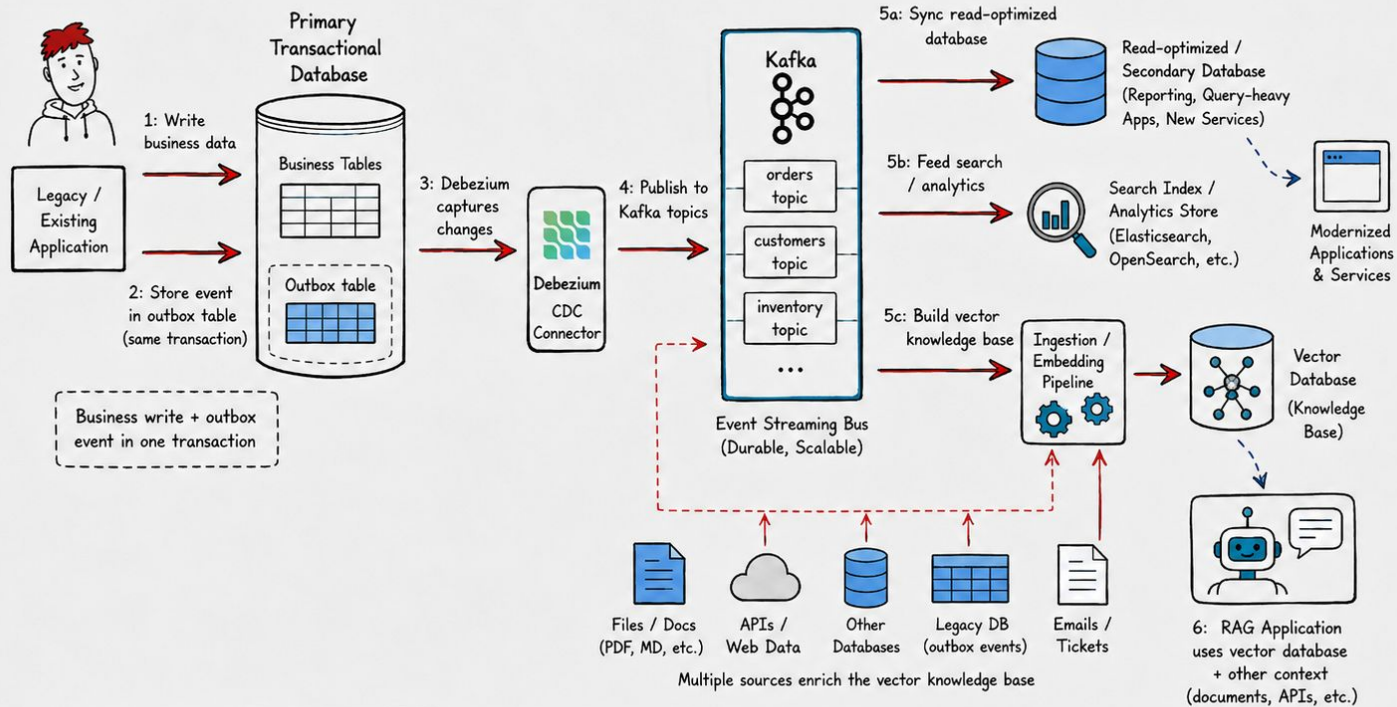


Why this helps AI / RAG

- Continuously updated knowledge
- Multiple sources in one knowledge base
- Better answers than model-only AI

So... What about the data? How to sync data sources?





Why this helps modernization:

- Decouple legacy (writes) from new apps (reads)
- Keep read workloads off the transactional DB
- Build a read-optimized database for new services
- Reuse events for multiple consumers
- Evolve independently without risky tight coupling

Why this helps AI / RAG:

- Continuously synced, up-to-date knowledge
- Multiple data sources for richer context
- Vector database built from many inputs
- Better answers than the model alone
- Event-driven updates keep knowledge fresh

- ✓ Safe modernization without direct coupling to legacy database
- ✓ Scalable, reliable, and observable pipeline
- ✓ The AI experience depends on synced data & knowledge, not just the model



Be lazy, shift

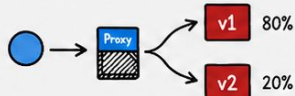
*Security, zero-trust,
certificates, monitoring,
observability, resiliency,
deployment patterns, ...*

Arguments to reduce the cognitive load
in agent development



1 Traffic & routing control

- Retries & timeouts
- Traffic splitting
- Header / path routing
- Mirroring

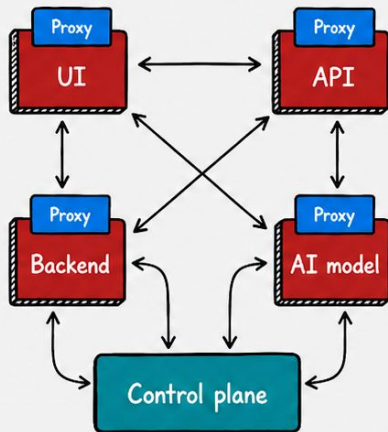


2 Secure communication

- mTLS between services
- Central certificate management
- Certificate rotation
- Workload identity



Service Mesh



Security



Observability



Traffic management



Policy

3 Safer deployments

- Canary releases
- Blue / Green
- Shadow traffic
- Safer AI model rollouts



4 Observability & monitoring

- Metrics
- Logs
- Traces
- Service map



Why it matters



Centralized policy enforcement



Better visibility and troubleshooting



Reduced deployment risk



More resilient modern applications

So.... What is that service mesh about?



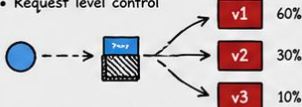
1 Centralized routing policies

- Retries
- Timeouts
- Circuit breaking
- Fault handling
- Policy consistency across services



2 Advanced routing policies

- Path or header based routing
- User or tenant specific routing
- Traffic splitting
- Mirroring
- Request level control



4 Centralized certificate management



- mTLS between services
- Certificate issuance and rotation
- Workload identity
- Encrypted service-to-service traffic

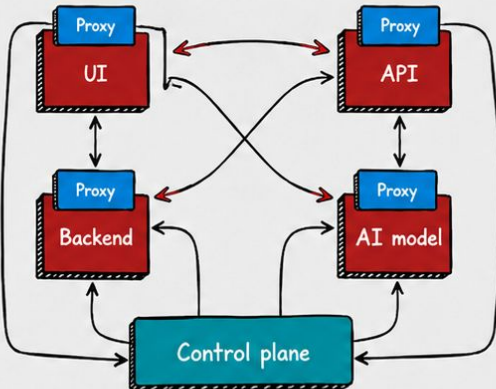


6 Why this matters for AI and modern apps



- ★ Consistent, secure communication between UI, API, backend, and AI model
- ★ Safer rollout of AI model and prompt changes (canary, mirroring)
- ★ Easier governance and policy enforcement across all services
- ★ Better reliability, performance, and visibility
- ★ Reduced operational risk in complex, distributed systems

Service Mesh



Security



Observability



Traffic management



Policy enforcement



Policies (YAML)



Identity Provider



Certificates (Issuance & Rotation)



Metrics & Telemetry Sources



Configuration & APIs

3 Advanced deployment patterns

Canary releases



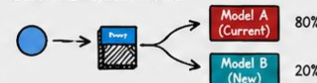
Blue/Green deployments



Mirroring / Shadow traffic



Safer AI model rollouts



5 Observability & monitoring



Metrics



Logs



Traces



Service Map



Latency



Errors



End-to-end insight across services

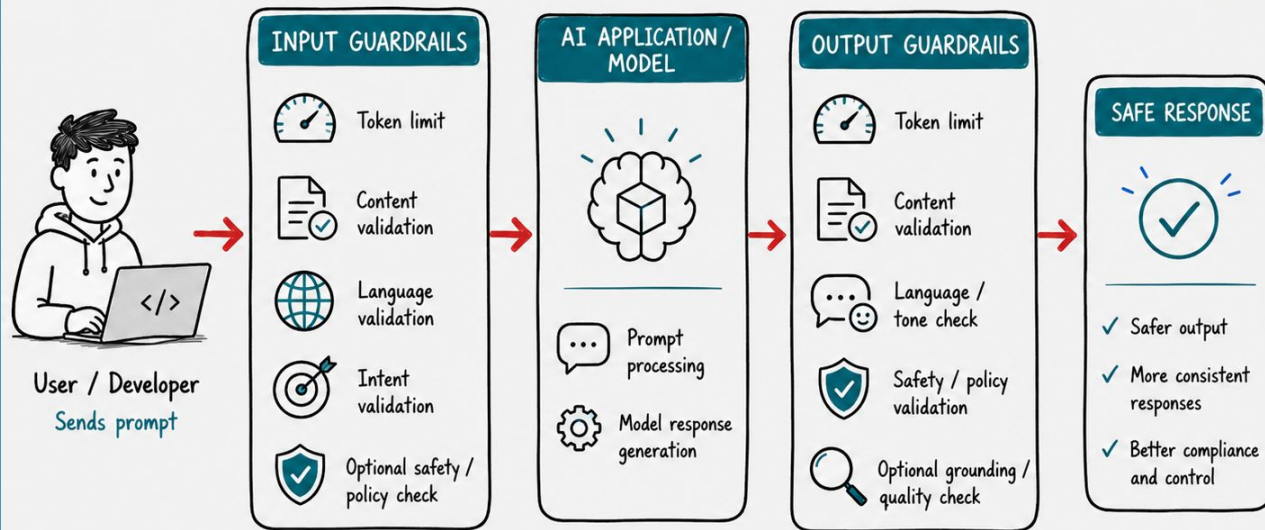
Resulting value

- ✓ More control over how traffic flows
- ✓ More security with identity & encryption
- ✓ Safer deployments with less risk
- ✓ Better visibility and faster troubleshooting
- ✓ Happier users and more resilient apps



AI guardrails on input and output

Validate requests before the model, and validate answers before returning them.



Guardrails reduce risk by checking both what goes into the model and what comes out.

So.... Is service mesh enough to secure my AI?



Recap

- Challenges:
Cognitive load, what about the data, SDLC/WDLCs
- # AI architectures/implementations
- "Just Calling AI"
- Deployment - responsibility
- What about the data?
- Service mesh & guardrails





*Developing agents or
developing with agents?*

Be lazy, shift

Arguments to reduce the cognitive load
in agent development



The idea you get at 2PM: It needs fine-tuning and (security) review

**Check recorded talks:
AI without spaghetti.**



Source: <https://drinkotron.com/memes/drinkotron-drunk-forty-40-years-hat-grandma-meme>

... Why the ~~xxx~~?
How is this
related to vibe
coding?

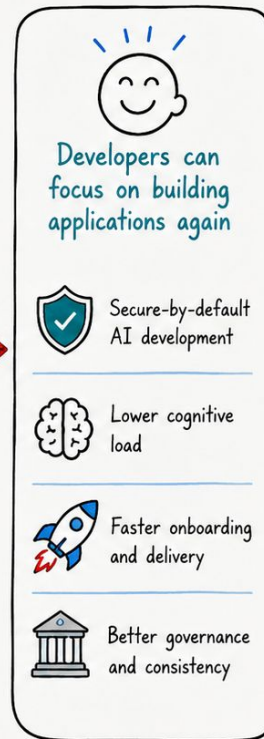
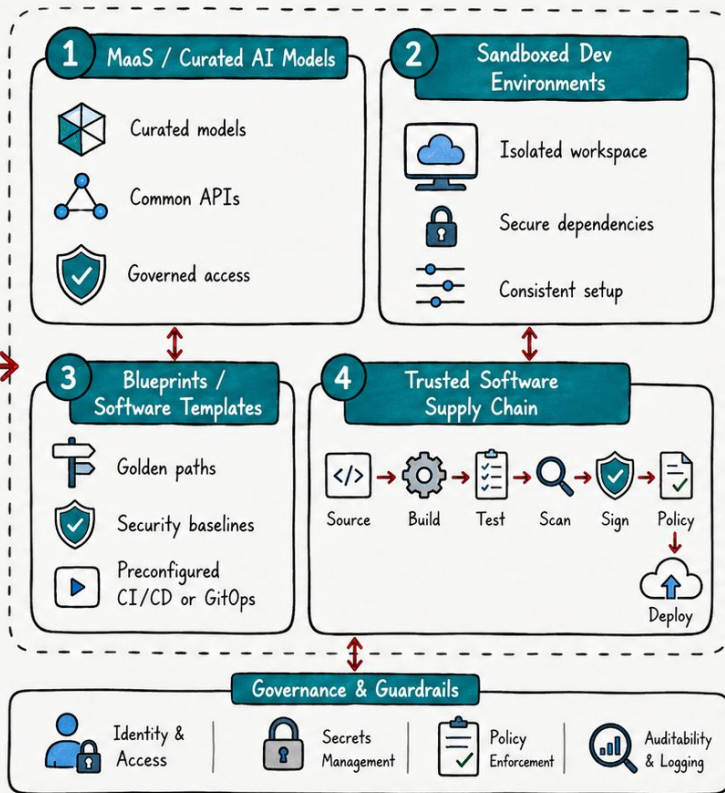




Developer



Sign in



Secure-by-default. Lower cognitive load. Faster delivery. Better governance. Developers can focus on building again.



So... Secure AI development environment



1 MaaS reduces cognitive load

- Curated models
- Common access patterns / APIs
- Rate limiting / governance
- Safer access to approved models
- Developers do not need to self-host every model



2 Sandboxed coding environments

- Isolated workspaces
- Secure dependencies
- Consistent developer setup
- Easier onboarding
- Reduced local machine complexity



3 Blueprints / software templates

- Golden paths
- Preconfigured security
- Standard CI/CD and GitOps setup
- Reusable architecture patterns
- Faster project bootstrap

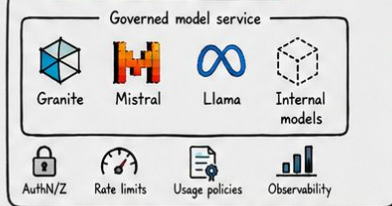


Sign in with company identity



Secure AI Development Environment

1. MaaS / Curated AI Models



2. Sandboxed Dev Environments

- ✓ Isolated workspaces
- ✓ Secure dependencies
- ✓ Ephemeral or persistent
- ✓ Preconfigured tools
- ✓ No direct internet egress

3. Blueprints / Templates

- ✓ App templates
- ✓ Architecture patterns
- ✓ Security baselines
- ✓ CI/CD & GitOps setup
- ✓ IaC modules

4. Trusted Software Supply Chain



5. Platform Guardrails & Governance



4 Trusted software supply chain

- Source control
- Build pipeline
- Tests
- Image scanning
- Signed artifacts
- SBOM / provenance
- Policy enforcement before promotion



5 Security & governance

- Identity / access control
- Secrets handling
- Policy checks
- Auditability
- Compliance support
- Approved tools and dependencies



6 Developers can develop again

- ✓ Less time on setup
- ✓ Less time on infra / security guesswork
- ✓ More focus on business logic and features
- ✓ Reduced cognitive load
- ✓ Better collaboration between platform teams and developers



Developer gets a secure workspace and templates



Workspace connects to governed MaaS endpoints



Code is built and validated through a trusted pipeline



Guardrails enforce policies before promotion



Approved artifact is deployed to trusted targets



Secure, standardized AI application delivery

Resulting value: developers can develop again



- ✓ Secure-by-default AI development
- ✓ Lower cognitive load for developers
- ✓ Faster onboarding and delivery
- ✓ Better governance and consistency
- ✓ Developers can focus on building applications again



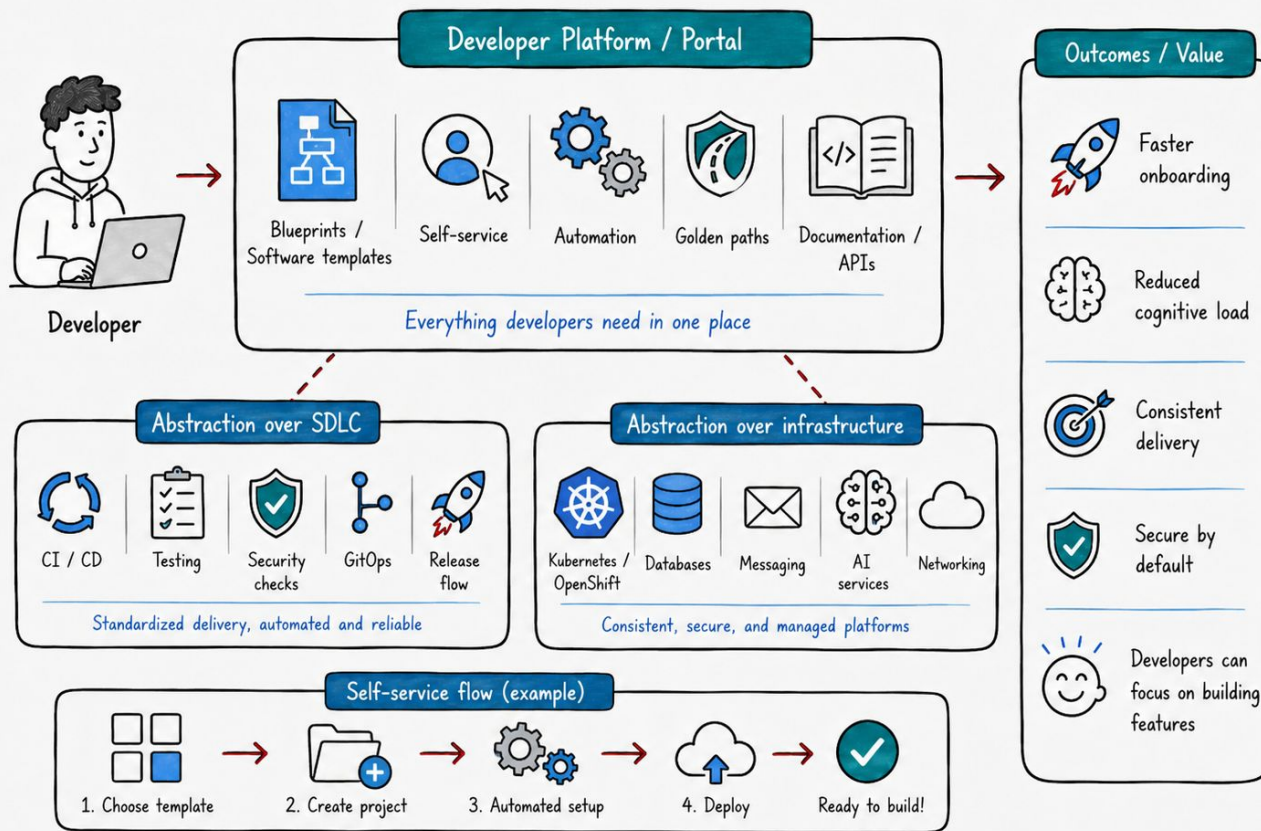
lazy, **shift right**

*Maarten, there are still
many, many things to do.
It's overwhelming*

reduce the cognitive load
development

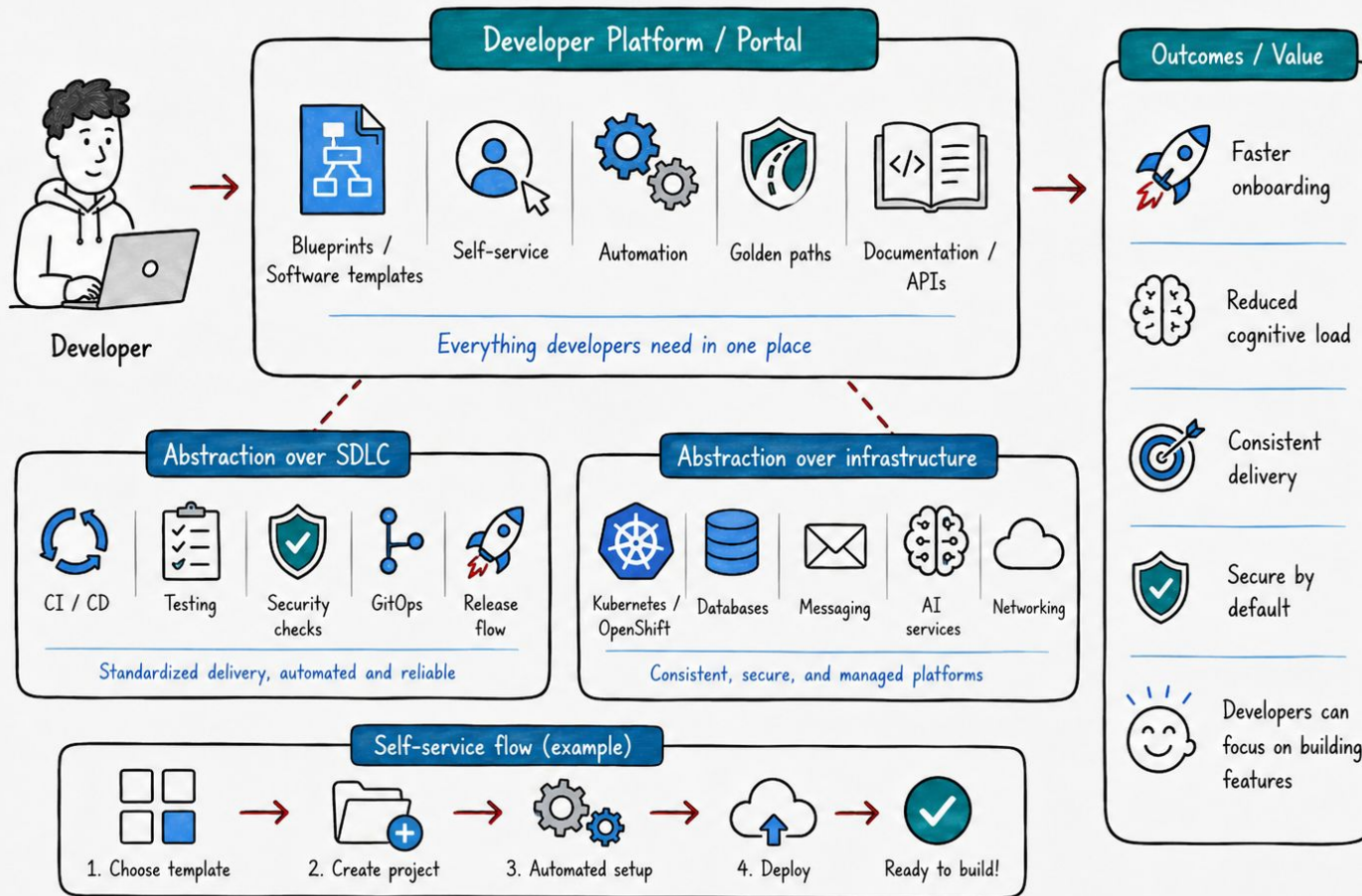


Check recorded talks: Platform Engineering 101.



How to reduce
the cognitive
load?





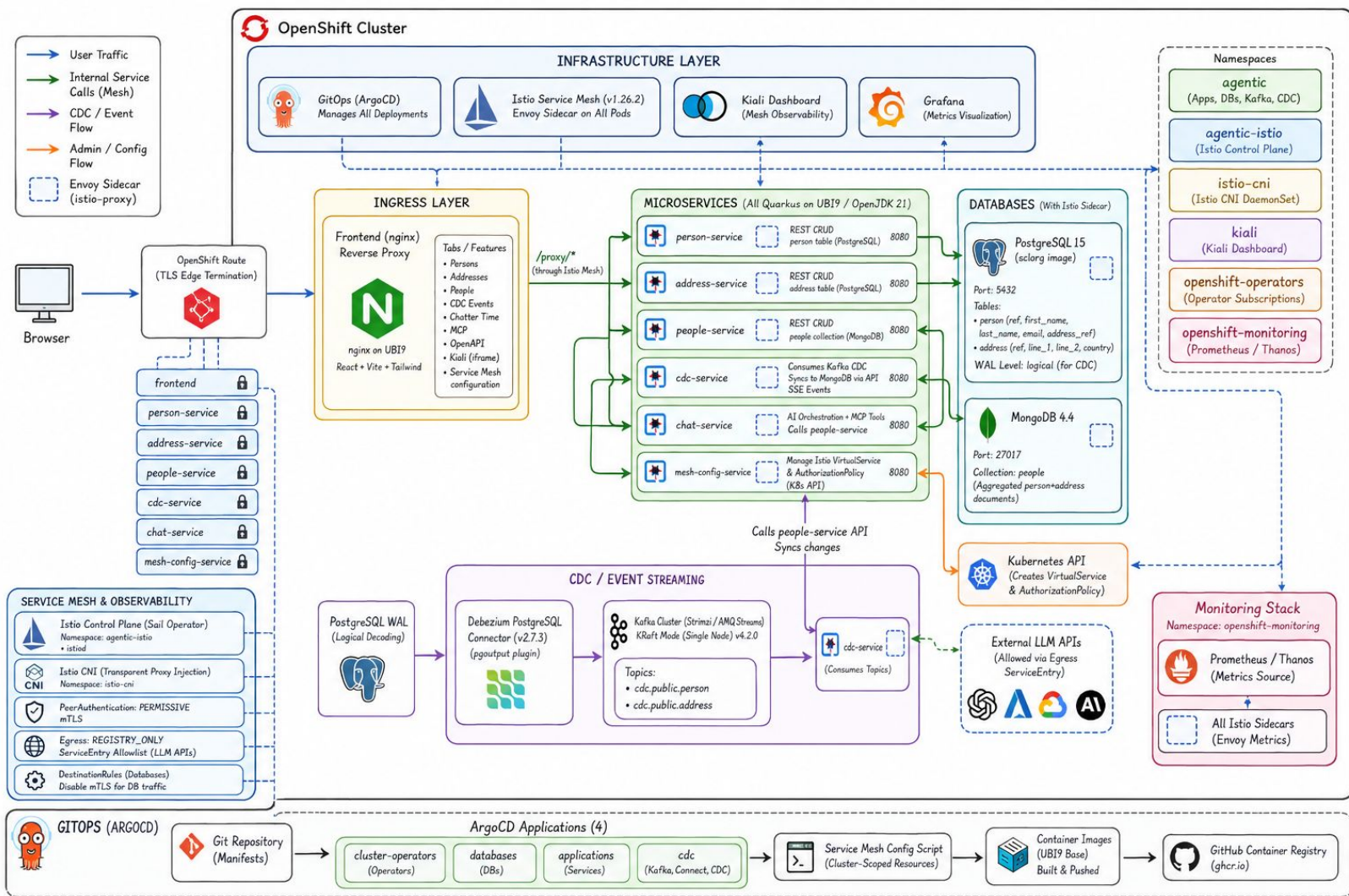
Recap

- Challenges:
Cognitive load, what about the data, SDLC/WDLCs
- # AI architectures/implementations
- "Just Calling AI"
- Deployment - responsibility
- What about the data?
- Service mesh & guardrails
- AI-infused development
- Platform engineering



The background of the image is a sunburst pattern. It consists of numerous thin, triangular rays of varying shades of blue, all radiating from a single point at the bottom center. The rays extend towards the top and sides of the frame, creating a sense of dynamic movement and energy.

DEMO TIME



Recap

- Challenges:
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- # AI architectures/implementations
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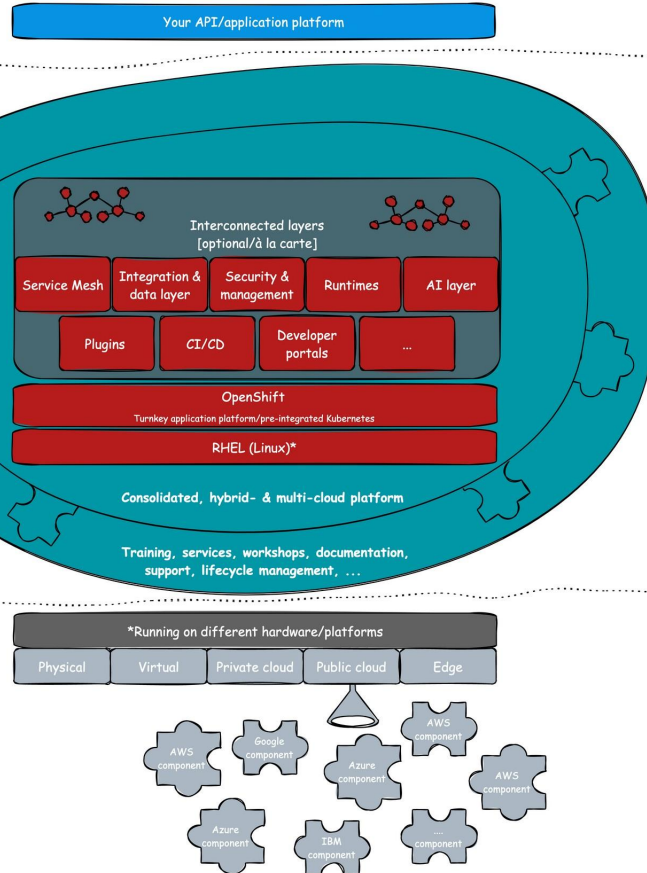


Applications

App & AI platform
⇒ software factory

Hybrid infrastructure

Red Hat



So... you deliver the
(enterprise-grade) building
blocks?
⇒ developers.redhat.com

*Be lazy, shift right.
Reduce the cognitive load
in (agent) development .*

**“Keep your
options open!”**

maarten-vandeperre.be
developers.redhat.com

